

YERSHOV, Russian Federation

WMO: 341860

Lat: 51.367N

Lon: 48.300E

Elev: 111

StdP: 100.00

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-22.8	-28.7	0.3	-25.4	-25.9	0.4	-22.5	12.1	-4.1	11.1	-4.9	3.0	20	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	35.0	20.0	33.0	19.5	31.0	18.8	21.3	31.1	20.5	29.9	19.7	28.6	4.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.4	23.8	17.4	12.6	23.4	16.5	11.9	23.2	62.2	31.3	59.3	29.9	56.5	28.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.3	8.3	DB	-28.4	37.1	3.4	1.8	-30.8	38.5	-32.8	39.5	-34.7	40.6	-37.1	41.9	(4)
(5)			WB	-28.6	22.8	3.3	1.5	-31.0	23.8	-32.9	24.7	-34.8	25.5	-37.2	26.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-10.1	-9.8	-3.4	8.0	15.6	20.6	22.9	21.5	14.7	6.8	-1.1	-7.4	(6)	
	DBStd	13.16	6.75	6.74	5.52	5.51	4.74	4.39	3.80	4.38	4.68	4.68	5.57	6.59	(7)	
	HDD10.0	2714	624	556	414	102	9	0	0	0	11	123	334	541	(8)	
	HDD18.3	4688	883	789	672	312	111	26	6	19	129	359	584	799	(9)	
	CDD10.0	1474	0	0	1	41	183	318	401	357	151	23	0	0	(10)	
	CDD18.3	406	0	0	0	1	27	94	148	117	19	0	0	0	(11)	
	CDH23.3	4417	0	0	0	17	288	999	1604	1288	219	2	0	0	(12)	
	CDH26.7	1878	0	0	0	1	74	423	726	592	61	0	0	0	(13)	
(14) Wind	WSAvg	4.1	4.5	4.8	4.6	4.4	4.1	3.6	3.4	3.5	3.6	3.9	4.2	4.4	(14)	
(15) Precipitation	PrecAvg	392	28	24	26	30	31	42	36	31	44	37	32	31	(15)	
	PrecMax	550	65	48	55	72	60	98	93	82	146	89	101	62	(16)	
	PrecMin	263	8	8	1	2	1	2	4	3	2	3	4	9	(17)	
	PrecStd	74	13	10	13	17	14	26	24	22	39	23	22	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	2.0	13.6	25.6	30.6	36.6	37.6	37.2	31.2	22.5	11.9	5.1	(19)	
		MCWB	2.5	1.0	8.7	15.0	18.3	19.8	19.9	20.3	17.1	13.6	9.6	4.1	(20)	
	2%	DB	1.1	1.2	8.5	22.5	28.8	33.9	34.9	34.6	28.4	19.3	9.4	2.5	(21)	
		MCWB	0.6	0.6	5.5	12.9	17.5	19.4	20.0	20.2	16.2	11.9	7.8	1.9	(22)	
	5%	DB	0.3	0.6	5.1	19.6	26.9	31.2	32.9	32.6	26.0	16.3	7.6	1.3	(23)	
		MCWB	-0.1	0.2	3.2	11.9	16.7	18.8	19.7	19.4	15.2	10.5	6.4	0.7	(24)	
	10%	DB	-0.5	-0.1	2.9	16.9	24.3	28.9	30.9	30.0	23.1	13.9	5.7	0.5	(25)	
		MCWB	-0.9	-0.4	1.5	10.5	15.4	18.3	19.0	18.5	14.2	9.5	4.7	0.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	1.4	8.6	16.1	20.4	21.6	22.2	23.3	18.7	15.0	10.5	4.5	(27)	
		MCDB	2.8	1.7	12.7	23.4	28.6	31.8	32.0	32.8	28.1	19.1	11.7	5.0	(28)	
	2%	WB	0.7	0.7	5.8	14.1	18.3	20.5	21.4	21.2	17.2	12.8	8.2	2.1	(29)	
		MCDB	0.9	1.0	8.0	20.3	26.9	30.1	31.1	31.2	25.2	17.5	9.1	2.4	(30)	
	5%	WB	0.0	0.2	3.4	12.4	17.1	19.7	20.6	20.0	16.0	11.4	6.5	0.9	(31)	
		MCDB	0.3	0.6	5.1	18.3	25.1	28.8	29.9	30.0	23.6	15.4	7.4	1.2	(32)	
	10%	WB	-0.9	-0.5	1.7	11.0	15.9	18.8	19.8	19.0	14.9	10.0	4.8	0.1	(33)	
		MCDB	-0.5	-0.1	2.7	16.8	23.1	27.4	28.5	28.4	21.8	13.3	5.7	0.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.6	6.9	9.8	11.1	11.3	11.4	11.8	11.0	8.4	5.4	5.5	(35)	
		MCDBR	4.2	4.5	8.7	13.7	13.7	14.2	14.0	14.4	15.2	12.7	6.7	4.0	(36)	
		MCWBR	4.2	4.3	6.1	6.8	6.4	5.4	5.0	5.7	7.0	7.4	5.5	4.0	(37)	
	5% WB	MCDBR	4.2	4.4	8.3	12.4	12.2	12.3	12.4	13.3	13.2	11.1	5.9	3.8	(38)	
		MCWBR	4.2	4.3	6.0	6.7	6.4	5.1	4.9	5.8	6.9	7.2	5.3	3.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.307	0.334	0.372	0.436	0.409	0.394	0.407	0.426	0.376	0.351	0.330	0.306	(40)	
	taud		2.079	2.026	2.064	2.084	2.255	2.350	2.312	2.240	2.350	2.353	2.281	2.140	(41)	
	Ebn at Noon		666	756	806	798	845	862	843	802	805	750	654	603	(42)	
	Edh at Noon		80	112	131	146	129	118	121	124	99	81	67	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.03	2.13	3.43	4.53	5.90	6.42	6.26	5.31	3.64	1.99	0.87	0.61	(44)	
	RadStd		0.21	0.27	0.50	0.40	0.42	0.55	0.43	0.39	0.45	0.31	0.13	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.72	N/A	N/A	+1.32	N/A	N/A	N/A	N/A	+183	+110	(46)	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+1.55	N/A	N/A	N/A	N/A	+179	+110	(47)	

Nomenclature: See separate page